

Estimation Appendix

Paradise Planner - Finding Events and Budgeting slice

1. Slice Definition

Item	Current definition
Slice name	Finding Events and Budgeting
Why selected	Budgeting was a key need in interviews; events influence where travelers go and what they do.
User inputs	Minimum/maximum budget slider and event filters.
User-visible results	Event list and places/destinations informed by average budget.
Stored/read data	Average price information for destinations and event data pulled periodically.
External connections	Travel-related external systems/data sources needed for the slice.
Out of slice	Hotels, flights, weather, payments, finding travel companions, and AI features.

2. Stories and Backlog Size

User story	MoSCoW	Points
As an app owner, I want access to affordable event data so that I can provide it to users.	Must Have	8
As a prospective traveler, I want to find events so that I know where I should travel.	Must Have	5
As a prospective traveler, I want a good budgeting system so that I can travel comfortably without financial stress.	Must Have	8
As a prospective traveler, I want to see the average cost of visiting so that I can see where I can go.	Should Have	2
As prospective travelers, I want to find people so that we can hang out.	Won't Have	5
As a prospective traveler, I want to see bus prices so that I can plan in advance for traveling.	Could Have	8

Total backlog points recorded in the activity: 36. Note: this total includes lower-priority and Won't Have stories from the worksheet; the selected Sprint 2 estimates do not use story points as an input.

3. Method 1 - Function Points

Category	Count	Weight	Points
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Category	Count	Weight	Points
External Inputs (EI)	1	4	4
External Outputs (EO)	2	5	10
External Inquiries (EQ)	2	4	8
Internal Logical Files (ILF)	2	10	20
External Interface Files (EIF)	2	7	14
Total			56

Assumed productivity rate: 6 to 12 hours per function point. This produces $56 \times 6 = 336$ hours at the low end and $56 \times 12 = 672$ hours at the high end. A midpoint rate of 9 hours per function point gives a middle value of 504 hours.

4. Method 2 - Analogous Estimation

Input	Value / assumption
Comparable product	Wanderlog
Comparable team / first-release timeline	2 people, approximately 9 months (assumption recorded by the team)
Comparable effort	18 person-months
Fraction represented by our slice	10%-25%
Reason for the fraction	Wanderlog's early release focused on map/itinerary planning; events and budgeting were not the same scope. The upper end accounts for event-data integration complexity.
Team-size adjustment	1.15 multiplier (assumption)
Hours per person-month	160 hours (assumption)
Low estimate	$18 \times 0.10 \times 160 \times 1.15 = 331$ hours
Middle estimate	Approximately 580 hours using a 17.5% middle fraction
High estimate	$18 \times 0.25 \times 160 \times 1.15 = 828$ hours
Confidence	Low; the comparable timeline and fraction are judgment-based, and the comparable founders were experienced and full-time.

5. Team-Size Check

Check	Result	Explanation
Team size five or six at most	Yes	The working assumption is a four-person team.

Check	Result	Explanation
Communication paths counted	Yes	$4(4-1)/2 = 6$ communication paths.
Integration work considered	Yes	Integration is explicitly represented in the slice and budget assumptions.
No linear productivity assumption	Yes	The estimate does not assume doubling people doubles output.
Estimated for the team we have	Yes	The calculation uses a four-person team.
Comparable timeline adjusted rather than copied	Yes	The analogous estimate includes a 1.15 team-size adjustment.

6. Reconciliation of the Two Estimates

Measure	Function Points	Analogous
Range	336-672 h	331-828 h
Middle value	504 h	580 h
Primary driver	Hours per function point	Fraction of Wanderlog represented by the slice
Measured or assumed?	Assumed	Assumed

Gap between middle values: $580 / 504 = 1.15$. The estimates are relatively close despite using different starting information. The largest uncertainty is the assumed fraction of Wanderlog represented by the slice, followed by the assumed function-point productivity rate. More detailed design information and actual team productivity data would narrow the range.

7. Final Range and PERT Planning Value

The final range uses the lowest and highest values across both methods: O = 331 hours and P = 828 hours. The team selected M = 542 hours because it is the average of the two methods' middle values (504 and 580).

8. Cost and Calendar Conversion

Conversion	Result
Loaded rate	\$76 per productive hour (team assumption derived from \$85,000 salary, 1.4 burden factor, and 130 productive hours per FTE-month).
Labor-only range at \$76/hour	\$25,156-\$62,928
Calendar assumption	4 people x 30 productive hours per week = 120 team-hours/week
Calendar range	$331/120$ to $828/120$ = approximately 2.8-6.9 weeks

Conversion	Result
Expected labor-only cost	555 x \$76 = approximately \$42,180

9. Key Assumptions

- 6-12 hours per function point.
- Wanderlog comparable timeline and team size as recorded in the team research.
- The slice represents 10%-25% of the comparable first release.
- 1.15 team-size adjustment.
- 160 hours per person-month for analogous conversion.
- Four-person team and 30 productive hours per person per week.
- \$76 loaded productive-hour rate.
- Estimate date carried from the activity: October 4, 2026.